

Building your own DMR hotspot.

Parts.

While there are a number of places you can get the parts from the following list is specifically all Amazon links just because I know that these are from trustworthy suppliers, and these are the exact parts that I used in my build. These are **not** affiliate links so I make no money from your purchases there. If you choose to source your parts elsewhere, just be certain to get a Pi Zero 2 WH (With headers) for the Raspberry Pi, as it is much easier than trying to solder the headers on yourself, and it **must** be a Zero 2 as the original zero doesn't quite have enough horsepower for WPSD. Also if you want to use my suggested scripts for the UPS board, the only one I know for certain it works with is the SeenGreat one I have linked below.

1 – Pi Zero 2 WH Kit - <https://a.co/d/03xt3B0a>

1 – MMDVM Hotspot board in the Pi zero format with the OLED screen, Either Full Duplex or half, just have to make sure you print the correct “lid” when it comes time to do the 3d Printing. For **most** users the Half duplex board will suffice.

Full duplex - <https://a.co/d/04URRYBv>

Half Duplex - <https://a.co/d/07f9JcRm>

1 – PI Zero UPS hat from Seengreat - <https://a.co/d/0h6rxmVV>

1 – MicroSD card at least 16GB recommended – I recommend sandisk - <https://a.co/d/0hpMYj2s>

Total cost of all parts on Amazon as of this writing, around \$130.

Assemble all of the parts with the exception of the MicroSD card, putting the UPS board at the bottom of the stack with the contacts on it making contact with the bottom of the header pins on the Raspberry Pi itself. **Note:** on at least one build, I had to scrap some resin/varnish from the solder joints for those first six contacts on the bottom of the board in order to get the UPS board to make good contact. If switching on the UPS doesn't start lighting up LED's on the assembled boards, then take it back apart and try scraping at those contacts a bit. Note you will not be able to fully boot things up yet as you've not loaded the OS onto the card.

Additional Info and Downloads:

You'll want to download the latest version of WPSD for Raspberry Pi from here: [The WPSD Project - W0CHP.radio](#)

Additionally you'll need a utility to write that to the SD Card, I usually use Balena Etcher: [balenaEtcher - Flash OS images to SD cards & USB drives](#), but any similar utility should work.

Follow the instructions on the WPSD site to write the WPSD image to the card, then insert the card into the raspberry pi, plug a USB C power source into the UPS and switch the UPS on to start the boot up sequence. Note steps 3 -6 of the instructions on the WPSD site walk you through the initial boot and getting the WPSD system connected to wireless. Once you have that going, then you can go through initial setup. The following information should give you enough to get you going with Brandmeister. These instructions are specific for the half-duplex mmdvm board and assume that you already have an amateur radio license and call sign.

Initial WPSD configuration for Brandmeister:

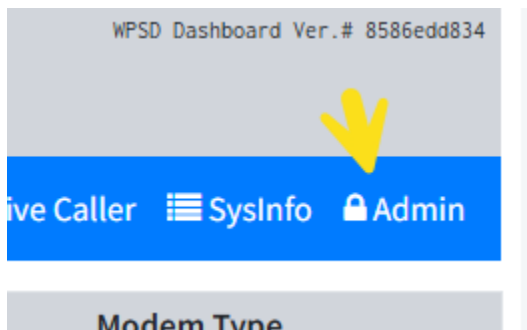
To complete the initial configuration, you will need a few things in advance.

1. A DMR id, which you can get at <https://radioid.net> signing up using your Call Sign.
2. An account at <https://Brandmeister.network> which again, you will create using your call sign. You will need the DRM radio ID *before* creating the Brandmeister account as it will need to know/be able to look up your DMR ID.
3. Determine what frequency you plan to use for your hotspot. In general I recommend using one of the frequencies recommended by the Colorado Council of Amateur Radio Clubs here: [CCARC Guidelines on Hotspots](#) even if you are not in Colorado, their guidance on using those frequencies still makes sense as far as channel spacing, and are in the frequency range that is typically recommended. For the purposes of this guide, I'll use 438.5000Mhz

With the Brandmeister account, you will need to two very specific pieces of information once you have your account there setup, login and navigate to the "Self Care" portal, which can be found by clicking on your call sign in the upper right corner and then selecting the self care button. On the self care page you'll need to set a HotSpot Security password. Enter your chosen password there, and click save, making sure you make a note of what that password is as you will need it to setup the hotspot. Then also navigate to your Profile screen, by again clicking your call sign in the upper right corner, and selecting "Profile". On

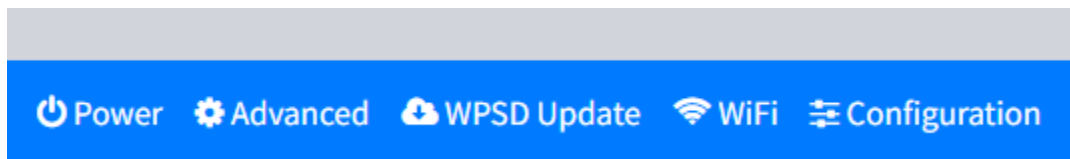
your profile page you will see a button under Security Settings for API Keys. Click that button and then click the Add button on the API key screen to generate a new API key, give it a name like “Hotspot API Key” and make sure to copy the API key out so you have that documented somewhere as well, you cannot view it again later.

Once you have gathered all the initial information and completed the initial Wifi configuration of your WPSD hotspot you are ready to proceed. From a device connected to the same wireless network as your new hotspot, open a web browser, and navigate to <http://wpsd.local> which should load the default homepage of the hotspot. From the dashboard click on the Admin link in the upper right corner, and you’ll be asked to log in.

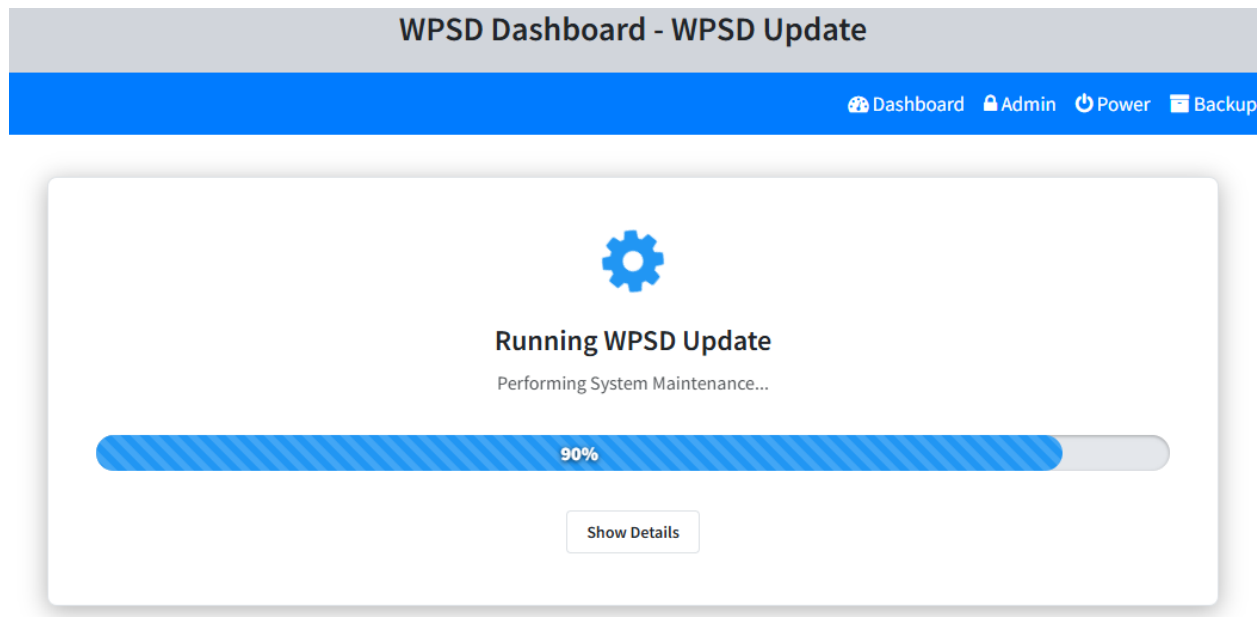


The default username is pi-star and the default password is raspberry.

The very first thing you should do before anything else, is go ahead and click that WPSD update button and make sure that it updates to the absolute latest version..



The update screen will show and run through updating to the latest and greatest version. Once that completes, you can then navigate to the Configuration option in the menu.



On the configuration screen we are going to focus on filling in the bare minimum to get your hotspot up and going on Brandmeister. After that you can go ahead and do additional research for the rest of the options on your own.

To start you can change the hostname if you would like, if this is your first hotspot, I would recommend leaving it alone, however if you have multiples then you might want to change this to append a number or something to the default of WPSD:

Hostname:	wpsd	Do not add suffixes such as ".local", etc. Note: A reboot is required for this change to take effect.
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In the Node callsign you ***must*** enter your assigned amateur radio call sign, as is, no additional suffixes, etc.

Node Callsign:	KD	Do not add suffixes such as "-G"
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In the DMR ID box you will fill in the DMR ID that you obtained from radioid.net earlier:

DMR/CCS7 ID:	3	Required for DMR Mode & DMR Cross-Modes (If you don't have one, get a DMR ID from Radioid.Net)
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You can leave the NXDN ID blank, and unless you bought a Duplex MMDVM board, you can leave the Radio Mode at the default of Simplex. Fill in the Radio Frequency as the one that you chose earlier from the CCARC guidelines, and select the MMDVM_HS_HAT for PI (GPIO) as the option for the Modem Type, leave the Modem Port and Baud rate alone.

Radio Mode:	☒ Simplex ☐ Duplex	Duplex mode requires Dual-Hat/Duplex Modems
Radio Frequency:	438.500.000 MHz	
Radio/Modem Type:	MMDVM_HS_Hat (DB9MAT & DF2ET) for Pi (GPIO)	
Modem Port:	/dev/ttyAMA0 Typically there is no need to manually change/set this; for advanced settings/usage.	
Modem Baud Rate:	115200	

Set the time zone and format to your desired settings, and the dashboard language to your desired language.

System Time Zone:	America/Chicago	Dashboard Time Format: ☐ 24 Hour ☒ 12 Hour
Dashboard Language:	english_us	

Under Node Location & Infor Settings, everything there is pretty much optional, but I do recommend filling in the correct information.

Next we're going to scroll down to the Radio/MMDVMHost Modem Configuration. Leaving everything else alone, turn on the DMR Mode and set the Primary DMR network to Brandmeister.

Radio/MMDVMHost Modem Configuration

Main Radio Modes				
D-Star Mode:	☐	RF Hangtime: 20	Net Hangtime: 20	
YSF Mode:	☐	RF Hangtime: 20	Net Hangtime: 20	
DMR Mode:	☒	RF Hangtime: 20	Net Hangtime: 20	Primary DMR Network: Brandmeister
P25 Mode:	☐	RF Hangtime: 20	Net Hangtime: 20	

Scroll down the MMDVMHost/Modem Display Configuration, and set the MMDVM Display type to OLED Type 3 (0.96" screen) the port Modem, and turn on the "Display always Active".

MMDVMHost/Modem Display Configuration

MMDVM Display Type:	OLED Type 3 (0.96" screen)	Port: modem
OLED Display Options:	Display Always Active: (Displays data even while modem is idle)	☒
	Scroll Display: (Note: OLED Type-3 [0.96"] displays only)	☐
	Rotate Display: (Rotates display orientation 180 deg.)	☐
	Invert Display: (Inverts display background/foreground)	☐

Next scroll down to DMR Configuration and we're going to configure the Brandmeister specific information. Select your BrandMeister Master from the list appropriate for your country. Enter the Hotspot Security password that you set earlier on the Brandmeister website. You will need to select the last two digits for your ESSID, you can select anything if

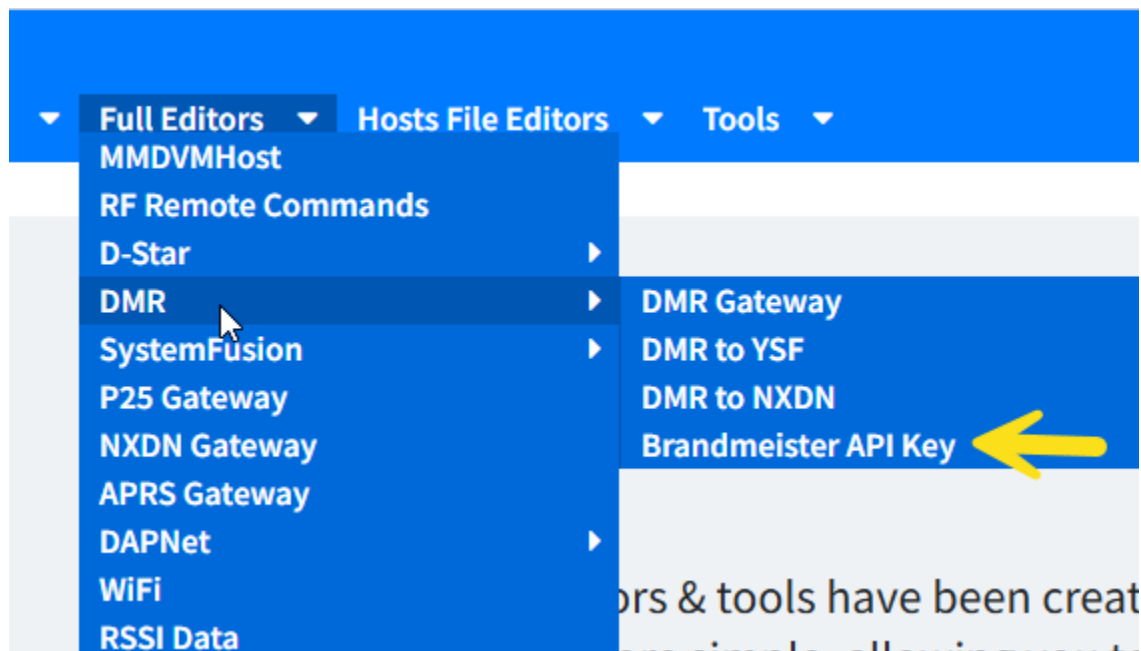
this is going to be your only hotspot. If you add other hotspots later, the key is to make sure each one has a unique suffix. Then set the slider to enable the BrandMeister network.

DMR Configuration

BrandMeister Network Settings	
BrandMeister Master:	BM_3102_United_States
BM Hotspot Security:	 Register for a Brandmeister Account...
BrandMeister Network ESSID:	03
BrandMeister Network Enable:	☒ Primary Network - No talkgroup prefix
BrandMeister Network:	Hotspot/Repeater Information Edit Hotspot/Repeater (BrandMeister Selfcare)
Brandmeister Manager:	BrandMeister Manager
DMR+ / FreeDMR / ADN / HBlind Network Settings	

For now you can leave everything else on the page alone, as you've been making changes, a banner will have popped down from the top of the page to apply your changes, you will also likely see a notification about creating a profile. WPSD allows you to create multiple Profiles so that you can switch between them, you can research more information about that on your own. For now simply click Apply Changes. It will take a few moments for the page to reload and the display on your board should light up now with some information.

Once the page has reloaded, click on the Advanced menu option at the top to go to the advanced configuration screen. Here you are going to select the menu option for Full Editors->DMR Configuration->Brandmeister API Key.



On that screen you will paste in your brandmeister api key that you retrieved earlier, and click apply changes.

[illegible]

Once you have done all of that your Hotspot screen should be showing the Powered by WPSD on the screen along with the WLAN0 IP address, and the CPU Temp. At this point your hotspot is ready for use. Program up some channels in your favorite DMR radio and give it a go. You should be able to program it as a simplex channel with your chosen frequency, Timeslot 2 and the color code of 1, picking your favorite DMR talkgroup on Brandmeister. If you've not used Brandmeister at all before, I recommend your first channel be for a private call on talkgroup 9990 which is the "parrot" talkgroup. You can use this to test your radio and ensure that it's getting out and receiving properly.

3d Printed case:

Plans for a 3d printed case can be found on Makerworld at <https://makerworld.com/en/models/2872935-case-for-dmr-half-duplex-pistar-with-ups-hat>

While that has a print profile directly for a Bambu Labs printer, you can download the 3mf file and should be able to print it on pretty much any 3d printer using your favorite slicer. The screw holes to screw the cover onto the rest of the case are designed for an M2x8mm screw. The whole stacked unit is a tight fit in that case, but should slide in fairly easily if you remove the SD card while assembling. The side where the UPS switch is located is a separate piece that you'll have to hold onto the side of the hotspot while sliding the whole thing down into the base.

Additional Configuration for the UPS

If you specifically utilized the UPS hat that I specified above, then you can do some additional configuration to have the system do a clean shutdown if it's running on the Battery from the UPS and the battery is starting to get low. For that you will need to be at least moderately comfortable with connecting to the raspberry Pi over SSH and adding/editing files there. The script has been updated to work with both WPSD and Pi-star at this point.

For the absolute easiest install you can see the instructions here:

<https://github.com/TheSystech/WPSD-SeenGreatUPS-Monitor>